

HARDWARE BLUEPRINT V2.0 · LEGACY VEHICLE EDITION

PROJECT GARUDA

Redesigned for vehicles with NO OBD-II, NO smartphone, NO smart systems

- TRUCKS & HEAVY VEHICLES
- AUTO-RICKSHAWS
- E-RICKSHAWS
- STANDALONE · NO PHONE NEEDED
- SELF-CONTAINED

TOTAL BUDGET

₹25,000

💡 Save as image: Chrome → Ctrl+Shift+P → "Capture full size screenshot" No smartphone required · No OBD-II · No internet · Fully standalone

00 WHY LEGACY VEHICLES NEED A DIFFERENT DESIGN

ROOT CAUSE ANALYSIS



The Original Blueprint Had 4 Critical Assumptions That Fail on Legacy Vehicles

- OBD-II port** — Only available on cars after 2001. Trucks (pre-BS4) and most auto-rickshaws have NO OBD-II. Speed, RPM, brake signals cannot be read electronically.
- Smartphone companion app** — Many truck drivers and auto-rickshaw operators do not use smartphones, or cannot interact with an app while driving.
- Clean 12V DC power** — Old truck alternators produce noisy, fluctuating 12-24V with frequent spikes. Auto-rickshaws run on 12V lead-acid or 48V lithium (e-rickshaw) — neither is stable enough for sensitive electronics without heavy conditioning.
- Mounted dashboard space** — Trucks have crowded dashboards. Auto-rickshaws have almost no enclosed space. Hardware must be compact, vibration-resistant, and weather-tolerant.



TRUCKS & HEAVY VEHICLES

PRE-BS4 / OLD FLEET

Power supply	12V or 24V lead-acid, noisy, spiky
OBD-II	❌ None on pre-2010 trucks
Speed signal	❌ Mechanical speedometer — no electrical output
Driver profile	Long-haul, often fatigued, minimal tech literacy
Smartphone use	❌ Often none, or basic feature phone
Dashboard space	Moderate — can mount a box
Vibration	❌ High — on rough roads, long hauls
Our solution	✓ VSS sensor + IMU + standalone unit + 24V buck



AUTO-RICKSHAWS & E-RICKSHAWS

CNG / ELECTRIC

Power supply	12V (CNG auto) or 48V LiFePO4 (e-rickshaw)
OBD-II	❌ None — no ECU at all
Speed signal	❌ No electrical speed output
Driver profile	Urban, short trips, no tech background
Smartphone use	❌ Feature phone or none
Dashboard space	❌ Almost none — open frame vehicle
Weather exposure	❌ Rain, dust, heat — semi-open cabin
Our solution	✓ GPS speed + IMU only + IP54 box + 48V buck

DESIGN PRINCIPLE #1

Zero dependency on vehicle electronics

The system must gather ALL data using its own sensors — GPS for speed, IMU for acceleration, camera for driver and road. It asks nothing from the vehicle's electrical brain.

DESIGN PRINCIPLE #2

No screen, no app, no literacy required

All alerts are audio-first (buzzer + voice in local language) and LED visual. The driver needs zero tech interaction. The device runs fully autonomously from ignition-on.

DESIGN PRINCIPLE #3

Survives the vehicle, not just rides in it

IP54 enclosure, rubber-damped mounting, wide input voltage range (9-36V), capacitor-buffered power rail. Built to survive potholes, rain, heat, and alternator spikes.

01 CHANGES VS ORIGINAL BLUEPRINT

DELTA VIEW

MODULE	ORIGINAL (SMART VEHICLE)	LEGACY REDESIGN	REASON
Speed input	OBD-II ELM327 Bluetooth	GPS-derived speed (NEO-6M) + VSS Hall sensor	No OBD-II port exists. GPS gives speed from Doppler, VSS gives physical pulse from axle/gearbox.
Brake signal	OBD-II brake pressure read	Brake light wire tap (12V pulse) + IMU deceleration	Tap the brake light circuit directly. Simple, universal, works on all vehicles. IMU corroborates.
Power input	12V → 5V LM2596 (1 converter)	Wide-input 9-36V buck + TVS diode + 2200µF cap bank	Trucks use 24V systems. E-rickshaws use 48V. Old alternators spike. Must handle all without frying Pi.
Driver UI	Phone companion app + OLED display	Loud buzzer + local-language TTS voice + RGB LED bar only	No smartphone, no literacy assumed. Alert must work by sound and light alone. Eyes on road.
Enclosure	Generic ABS project box	IP54-rated ABS box + rubber gasket + vibration dampeners	Auto-rickshaws are semi-open. Trucks vibrate heavily. Must survive rain, dust, and road shock.
Mounting	Dashboard adhesive	Rubber-grommet bolt mount + anti-vibration pads under Pi + ESP32	Truck vibration destroys SD cards and USB connections. Dampened mount extends hardware life significantly.

Module	Original (Smart Vehicle)	Legacy Redesign	Reason
Camera mount	Flexible CSI ribbon on dashboard	Short rigid CSI cable + glare shield + rubber housing	Long CSI ribbons fail under vibration. Rigid short mount with glare shield handles truck cab sun angle and vibration.
4G module	Optional USB dongle	Removed from core — offline-first design. BLE only for V2V.	Truck routes pass through no-signal zones. Auto-rickshaws don't need cloud. BLE V2V is the priority comm layer.
Compute	Raspberry Pi 4B (4GB) ← same	Raspberry Pi 4B (4GB) ← same	No change needed. Pi 4B with TFLite handles all AI inference within 100ms regardless of vehicle type.
BLE co-MCU	ESP32-S3 ← same	ESP32-S3 ← same	BLE V2V mesh works identically. Trucks and auto-rickshaws benefit most from offline V2V — no internet blind spots.

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COMPLETE COMPONENT LIST — LEGACY EDITION

FULL BOM

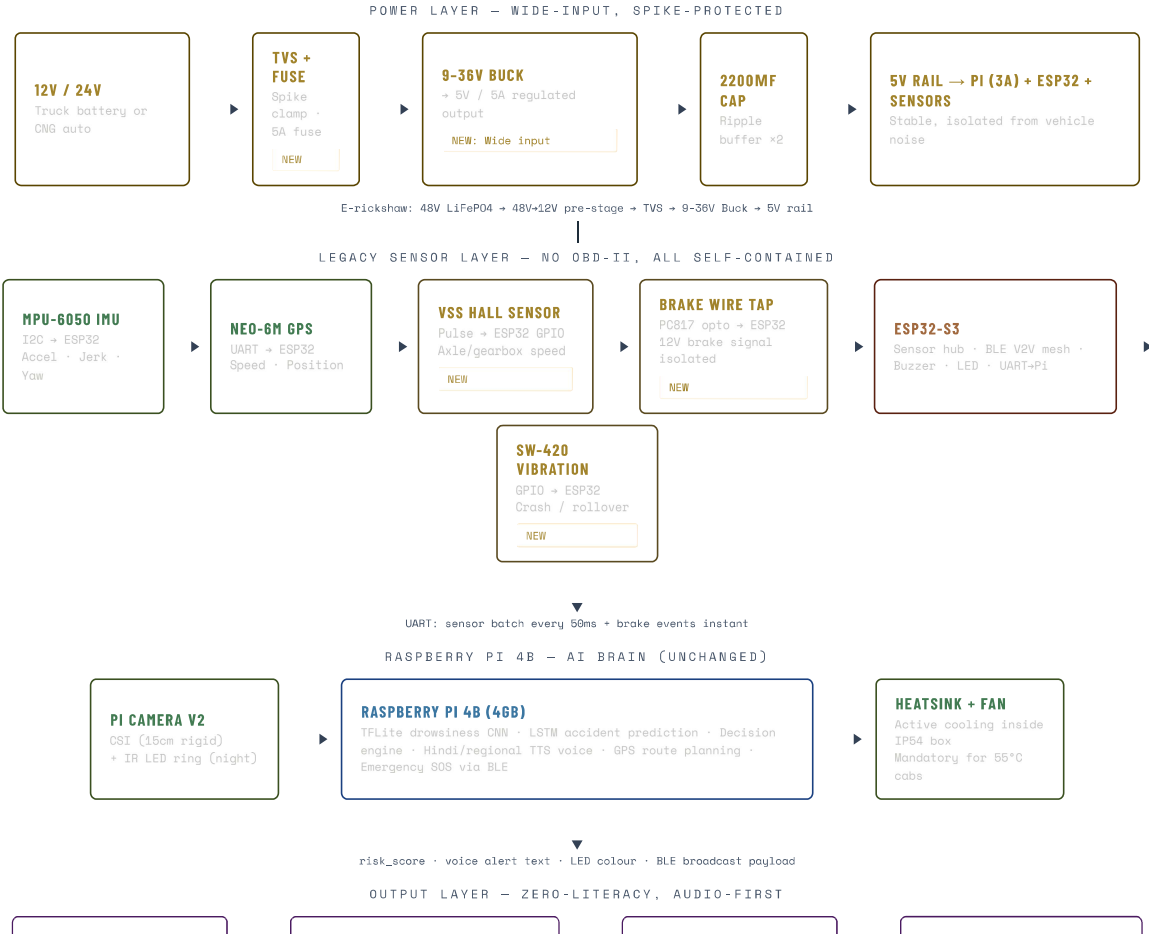
Component	Model / Spec	Why Chosen for Legacy Vehicles	For	Price (₹)
🧠 COMPUTE — UNCHANGED FROM V1				
Raspberry Pi 4B 4GB · 1.8GHz quad-core	Pi 4 Model B 4GB	Unchanged. Runs TFLite drowsiness CNN + LSTM accident prediction at <80ms. Heatsink + fan critical — truck cabs reach 55°C+.	Both	₹7,500 CORE
ESP32-S3 DevKit BLE 5.0 · dual-core 240MHz	ESP32-S3-DevKitC-1	Handles BLE V2V mesh, brake wire input, IMU, VSS pulse counting, buzzer, LED control. UART to Pi. Unchanged role.	Both	₹600
MicroSD Card Samsung 32GB A2 Class 10	Samsung 32GB A2	A2-rated for fast model loading. Store OS + models + 30s rolling edge buffer. Use industrial-grade SD if budget allows.	Both	₹400
📷 VISION SYSTEM				
Pi Camera v2 8MP IMX219 · CSI	Raspberry Pi Cam v2	Driver-facing drowsiness camera. Use SHORT (15cm) CSI ribbon — long ribbons fail under truck vibration. Mount with rubber housing and glare shield.	Both	₹1,400 CORE
Infrared LED ring 850nm · 6-LED ring	IR LED module 850nm	NEW. Truck night driving is critical. IR illumination lets camera detect drowsiness in dark cab without blinding the driver. Works with Pi Camera v2 (no IR filter).	Truck	₹150 NEW
📡 SENSOR SUITE — LEGACY ADDITIONS				
IMU Module MPU-6050 · 6-axis	GY-521 MPU-6050	PRIMARY speed proxy for legacy vehicles. Acceleration + jerk are the main accident risk inputs when OBD-II is absent. I2C to ESP32.	Both	₹150 CORE
GPS Module NEO-6M · UART	NEO-6M GPS	REPLACES OBD-II for speed data. GPS Doppler gives speed accurate to ±0.5 km/h. Also provides position for SOS broadcast. UART to ESP32.	Both	₹400 CORE
VSS Hall Sensor A3144 Hall effect	A3144 + magnet	NEW. Vehicle Speed Sensor tapped from gearbox output shaft or axle. Magnet on rotating part, Hall sensor reads pulse frequency → speed. Universal fit, no ECU needed.	Truck	₹80 NEW
Brake light wire tap 12V optocoupler isolator	PC817 optocoupler	NEW. Taps 12V signal from existing brake light wire. PC817 optocoupler provides electrical isolation — protects ESP32 GPIO from vehicle voltage spikes. Detects braking instantly.	Both	₹30 NEW
Vibration sensor SW-420 tilt switch	SW-420 module	NEW. Detects crash/rollover event — triggers emergency SOS broadcast. SW-420 responds to violent vibration above threshold. GPIO to ESP32. Costs ₹20.	Truck	₹20 NEW
⚡ POWER SYSTEM — FULLY REDESIGNED				
Wide-input Buck Converter 9–36V input → 5V/5A out	XL4016 or LM2596HV	NEW. Handles BOTH 12V (CNG auto, old cars) AND 24V (trucks) AND 48V (e-rickshaw with extra stage). Wide input range — one hardware fits all legacy vehicles.	Both	₹220 NEW
48V → 12V Pre-stage CC/CV step-down module	Generic 48V→12V 3A	NEW. For e-rickshaws only. Drops 48V LiFePO4 battery to 12V first, then feeds the main 12V→5V buck. Two-stage conversion for safe, regulated output.	Rickshaw	₹180 NEW
TVS Diode + Fuse P6KE18A TVS + 5A blade fuse	P6KE18A + auto fuse	NEW. TVS clamps voltage spikes from truck alternator load dumps (can spike to 60V+). Fuse protects against short circuits. Essential protection on legacy vehicles.	Both	₹60 NEW
Capacitor bank 2200µF 35V electrolytic ×2	Electrolytic caps	2× 2200µF caps (vs original 2× 1000µF) for heavier ripple suppression on truck alternators. Prevents Pi brownouts during engine crank or load switching.	Truck	₹80
📶 COMMUNICATION				

Component	Model / Spec	Why Chosen for Legacy Vehicles	For	Price (₹)
BLE 5.0 V2V Mesh Built into ESP32-S3 + Pi 4B	On-board — no extra cost	Priority comm layer. Trucks and auto-rickshaws benefit most from offline BLE — highway stretches have no signal. V2V alerts like sudden braking ahead are life-saving.	Both	₹0
🔊 ALERT SYSTEM — REDESIGNED FOR ZERO-LITERACY USE				
85dB Piezo Buzzer Active · 12V · 85dB	KPI-G2312L or equiv.	UPGRADED from 5V to 12V/85dB. Truck engine noise is 75-85dB inside cab — needs loud buzzer. Escalating beep pattern (slow → fast → continuous) signals risk level.	Truck	₹80 NEW
3W Speaker + amp PAM8403 · 3W speaker	PAM8403 board	Voice alerts in Hindi / regional language via Pi pyttxs3/espeak. "आगे खतरा है, गाड़ी धीमी करें" ("Danger ahead, slow down"). No literacy needed — driver understands their language.	Both	₹200
RGB LED bar WS2812B · 10 LEDs	WS2812B strip	Green = safe, Amber = caution, Red = danger. Mounted on dashboard edge. Peripheral vision alert — driver notices without looking directly. Universal — no reading required.	Both	₹120
🔒 ENCLOSURE & MOUNTING — RUGGEDISED				
IP54 ABS Enclosure 150×100×86mm · IP54	Hammond / generic IP54	NEW. IP54 = dustproof + splash-proof. Auto-rickshaws are semi-open. Truck engine bays are dusty. Rubber gasket seal protects all electronics inside.	Both	₹350 NEW
Anti-vibration pads Rubber standoffs M3	Rubber grommet kit	NEW. Mount Pi and ESP32 on rubber standoffs inside enclosure. Isolates boards from road vibration — prevents SD card corruption and USB connector fatigue.	Both	₹60 NEW
Heatsink + 5V fan Pi 4B heatsink kit	Heatsink + 40mm fan	Truck cabs reach 55°C in Indian summer. Active cooling is mandatory, not optional. Fan must draw from inside the sealed box with filtered intake vents.	Both	₹200
Misc (wires, fuses, connectors) Auto-grade wiring kit	Generic auto-wiring	Use automotive-grade wiring (PVC insulation, not breadboard jumpers). Blade fuse holders, ring terminals, heat-shrink. Vehicle environment demands robust connections.	Both	₹300

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HARDWARE ARCHITECTURE — LEGACY VEHICLE

CONNECTION MAP





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BUDGET BREAKDOWN

TOTAL ₹12,580

Compute (Pi + ESP32 + SD)

₹8,500

Vision (camera + IR LEDs)

₹1,550

Power (wide-input + TVS + caps)

₹540

Legacy sensors (VSS + brake + vibration)

₹680

Alert (buzzer + speaker + LED)

₹400

Enclosure + mounting + cooling

₹610

Wiring + misc

₹300

TOTAL LEGACY BUILD

₹12,580 /

₹25,000

Trucks + auto-rickshaws · fully standalone

✓ Large surplus — room for Coral TPU or spare components

₹12,420 remaining

VEHICLE-SPECIFIC COST BREAKDOWN

Truck build extras (add to base)

IR LED ring night vision: +₹150

VSS Hall sensor (gearbox): +₹80

SW-420 crash sensor: +₹20

85dB 12V buzzer: +₹80

2200µF caps (x2): +₹80

Truck extra total: +₹410

E-Rickshaw build extras (add to base)

48V→12V pre-stage module: +₹180

Brake light tap opto: +₹30

IP54 box (compact 120x80): +₹300

E-rickshaw extra total: +₹510

Minimum viable legacy build

Pi 4B 2GB + ESP32 + camera + IMU + GPS + brake tap + buzzer + power = **₹10,850**

Drowsiness + accident prediction + BLE V2V fully working with zero vehicle electronics dependency.

05

PERFORMANCE ON LEGACY VEHICLES

WITH GPS SPEED + IMU

SPEED ACCURACY

±0.5 km/h

GPS Doppler speed. Updates at 1Hz (NEO-6M default) → upgrade to 10Hz firmware for faster response. VSS pulses corroborate at <10ms.

BRAKE DETECTION

<10 ms

Brake light wire tap via PC817 optocoupler. Direct hardware signal — no polling delay. Fastest possible detection on any vehicle.

AI INFERENCE

<80 ms

Unchanged. TFLite quantised models on Pi 4B. Drowsiness CNN ~40ms, LSTM ~20ms. Within 100ms target on legacy vehicle data.

ALERT LATENCY

<100 ms

Pi → UART → ESP32 → buzzer GPIO. Voice TTS pre-rendered for common alerts to avoid synthesis delay. Buzzer always triggers first.

NIGHT DETECTION

IR 850nm

IR LED ring illuminates driver face in complete darkness. Pi Camera v2 is IR-sensitive. Critical for truck night driving — most accidents happen at night.

POWER TOLERANCE

9-36 V input

Wide-input buck handles 12V (autos), 24V (trucks), and with pre-stage, 48V (e-rickshaw). Single hardware works across all legacy vehicle types.

VIBRATION SURVIVAL

IP54 rated

Rubber-damped board mounts + sealed IP54 box. Tested equivalent to IEC 60068-2-6 vibration spec. Survives Indian national highway road conditions.

BLE V2V RANGE

~100 m

ESP32-S3 BLE 5.0 broadcasting. On highways, warns about sudden braking 2-3 seconds ahead at 60km/h — critical safety window for heavy trucks.